

Applications of Operational Amplifiers

Linear Applications • Active Filters & Oscillators • Comparators & Converters

A complete study of op-amp based circuits used in analog signal processing and waveform generation



What We Will Cover

01

Linear Applications

Amplifiers, followers, summing/scaling/averaging, differential & instrumentation amps, V-I / I-V converters, integrator & differentiator

02

Active Filters & Oscillators

1st & 2nd order LPF/HPF, band-pass, band-reject, all-pass filters; phase-shift & Wien bridge oscillators; square, triangular & sawtooth generators

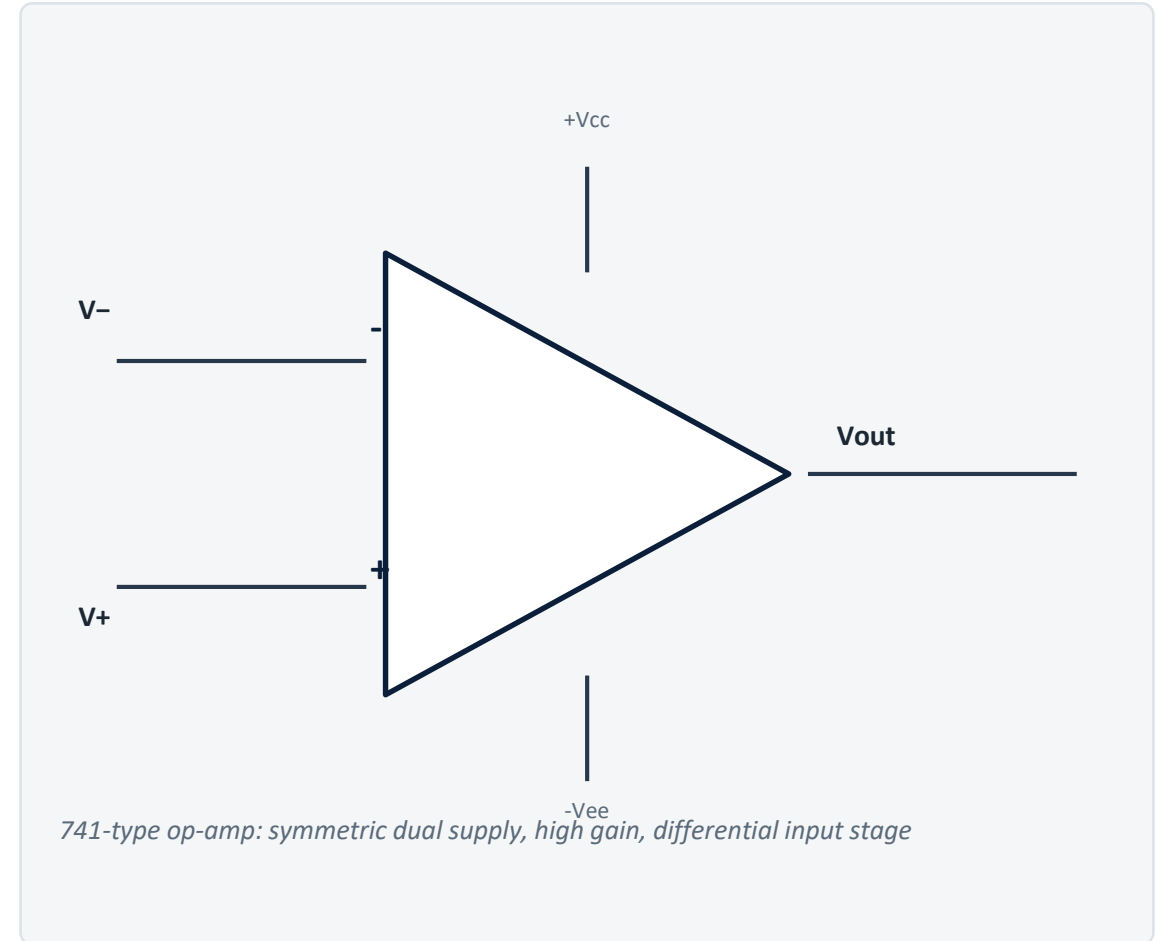
03

Comparators & Converters

Comparator, zero-crossing detector, Schmitt trigger, clippers & clampers, absolute-value circuit, peak detector, ADC & DAC

The Ideal Op-Amp: A Quick Recap

- Infinite open-loop gain ($A \rightarrow \infty$)
- Infinite input impedance — no current enters either input
- Zero output impedance — ideal voltage source at output
- Infinite bandwidth and zero input offset voltage
- Golden Rule 1: With negative feedback, $V_+ = V_-$ (virtual short)
- Golden Rule 2: No current flows into the input terminals





01

SECTION ONE

Linear Applications

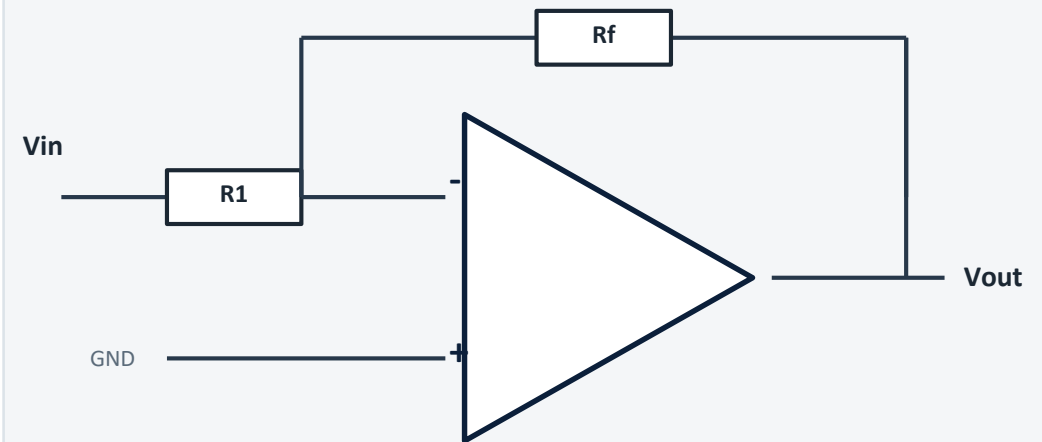
DC and AC Amplifier

- Inverting amplifier: input applied to (–) terminal through R1; feedback resistor Rf sets the gain
- Non-inverting amplifier: input applied directly to (+) terminal; output is in phase with input
- Coupling capacitors added at input/output convert a DC amplifier into an AC-only (capacitor-coupled) amplifier by blocking DC offset
- Gain is set purely by external resistors — independent of the op-amp's internal open-loop gain

GAIN EXPRESSIONS

Inverting: $A_v = -R_f/R_1$

Non-inverting: $A_v = 1 + (R_f/R_1)$



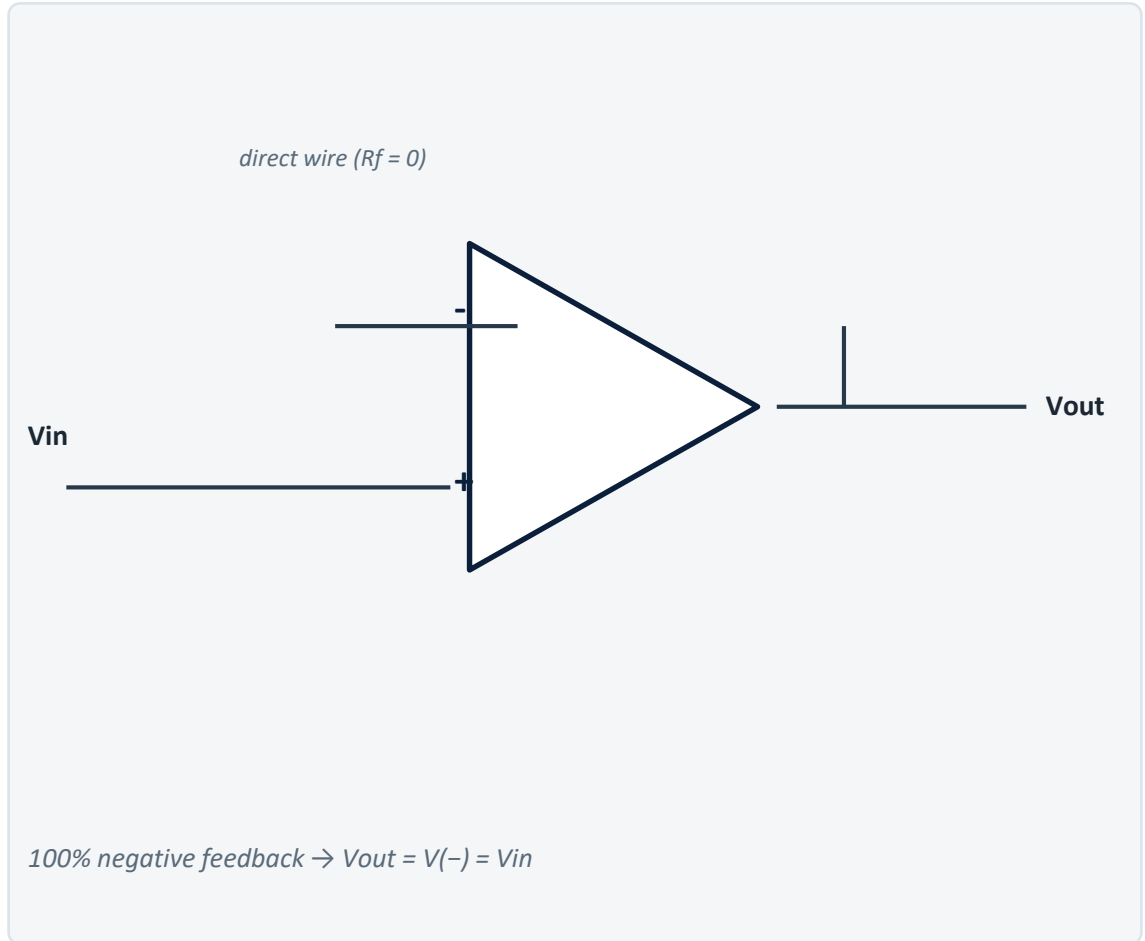
Inverting configuration shown — signal at (–) input, gain set by R_f / R_1

Voltage Follower (Buffer)

- Output is fed back directly (100%) to the inverting input — no external resistors needed
- Unity gain: V_{out} follows V_{in} exactly, $A_v = 1$
- Provides very high input impedance and very low output impedance
- Used as a buffer stage to isolate a high-impedance source from a low-impedance load, preventing loading effects

TRANSFER FUNCTION

$$V_{out} = V_{in} \quad (A_v = 1)$$

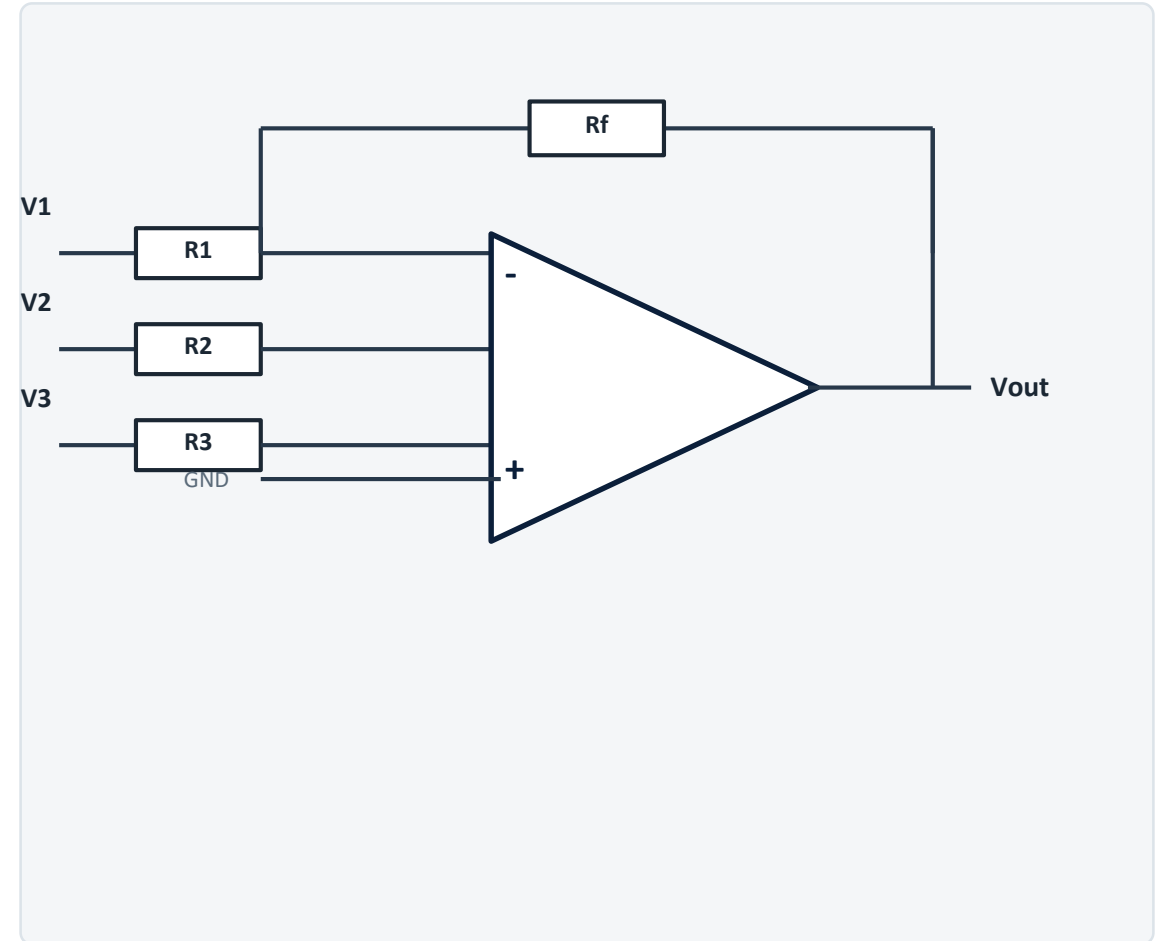


Summing Amplifier

- Multiple input signals, each through its own input resistor, are combined at the virtual-ground summing junction
- Because the (–) input is a virtual ground, currents from each input add algebraically at that node
- Output is proportional to the weighted sum of all inputs
- If all input resistors are equal ($R_1 = R_2 = R_3 = R$), the circuit becomes a simple adder

OUTPUT EXPRESSION

$$V_{out} = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$$



Scaling and Averaging Amplifier

- Scaling amplifier: a summing amplifier where each input resistor is chosen independently — every input gets its own weighting factor R_f/R_n
- Used to combine signals with different, deliberately chosen gains (e.g., audio mixing, weighted sensor combination)
- Averaging amplifier: a special case where all input resistors are equal ($R_1 = R_2 = \dots = R_n = R$) and $R_f = R/n$
- Output becomes the arithmetic mean of the input voltages
- Both circuits use the same summing-junction principle as the summing amplifier, just with different resistor ratios

SCALING

$$V_{out} = -(R_f/R_1 \cdot V_1 + R_f/R_2 \cdot V_2 + \dots)$$

AVERAGING ($R_1=R_2=R_n=R$, $R_f=R/n$)

$$V_{out} = -(V_1 + V_2 + \dots + V_n) / n$$

Design tip

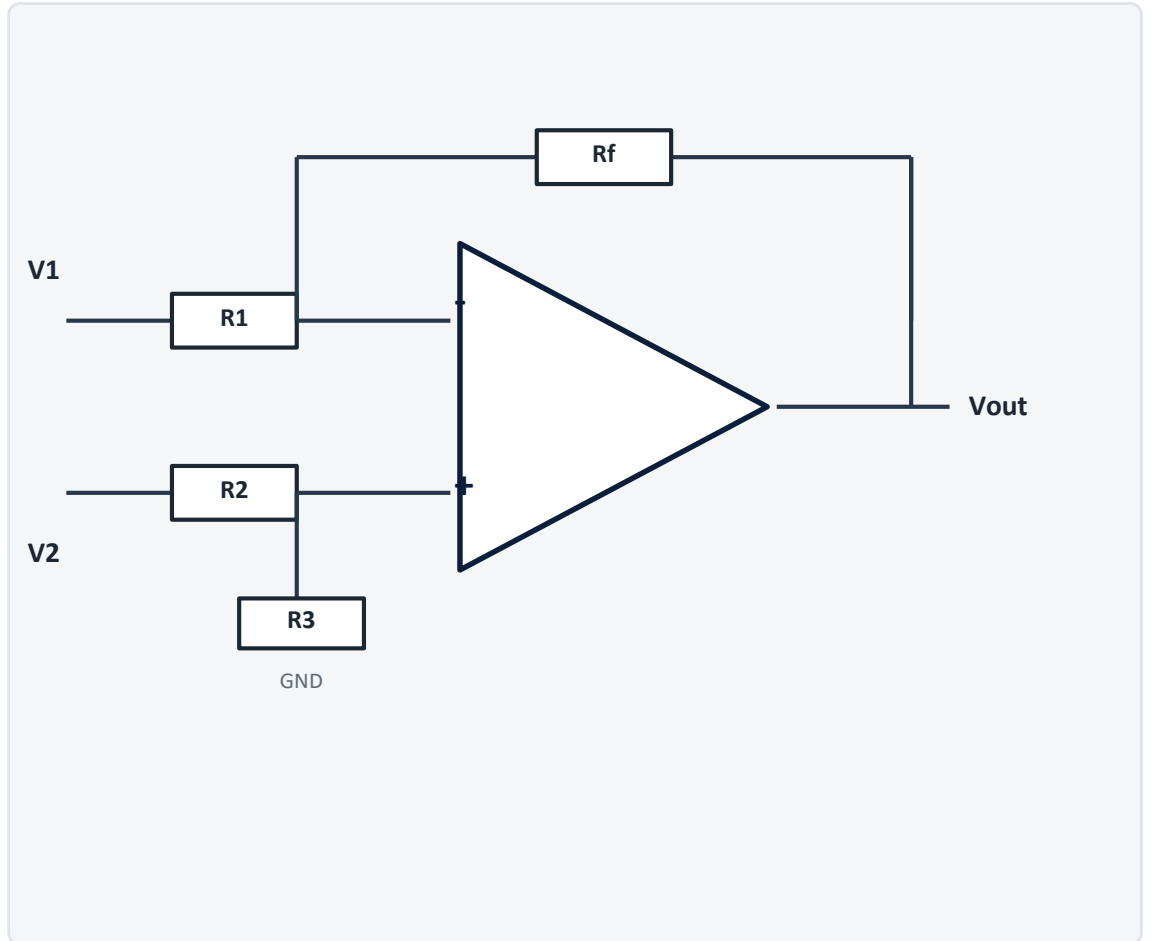
Averaging with $n = 3$ inputs: pick $R_1=R_2=R_3=30k\Omega$ and $R_f=10k\Omega$ so that $V_{out} = -(V_1+V_2+V_3)/3$.

Differential Amplifier

- Amplifies the difference between two input signals while rejecting any voltage common to both (common-mode rejection)
- Input V1 is applied through R1 to the inverting terminal; input V2 is applied through R2/R3 divider to the non-inverting terminal
- When $R1 = R2$ and $Rf = R3$, the circuit gives a pure difference output
- Key figure of merit: CMRR (Common Mode Rejection Ratio) — how well common signals are cancelled

OUTPUT ($R1=R2$, $Rf=R3$)

$$V_{out} = (Rf/R1)(V2 - V1)$$

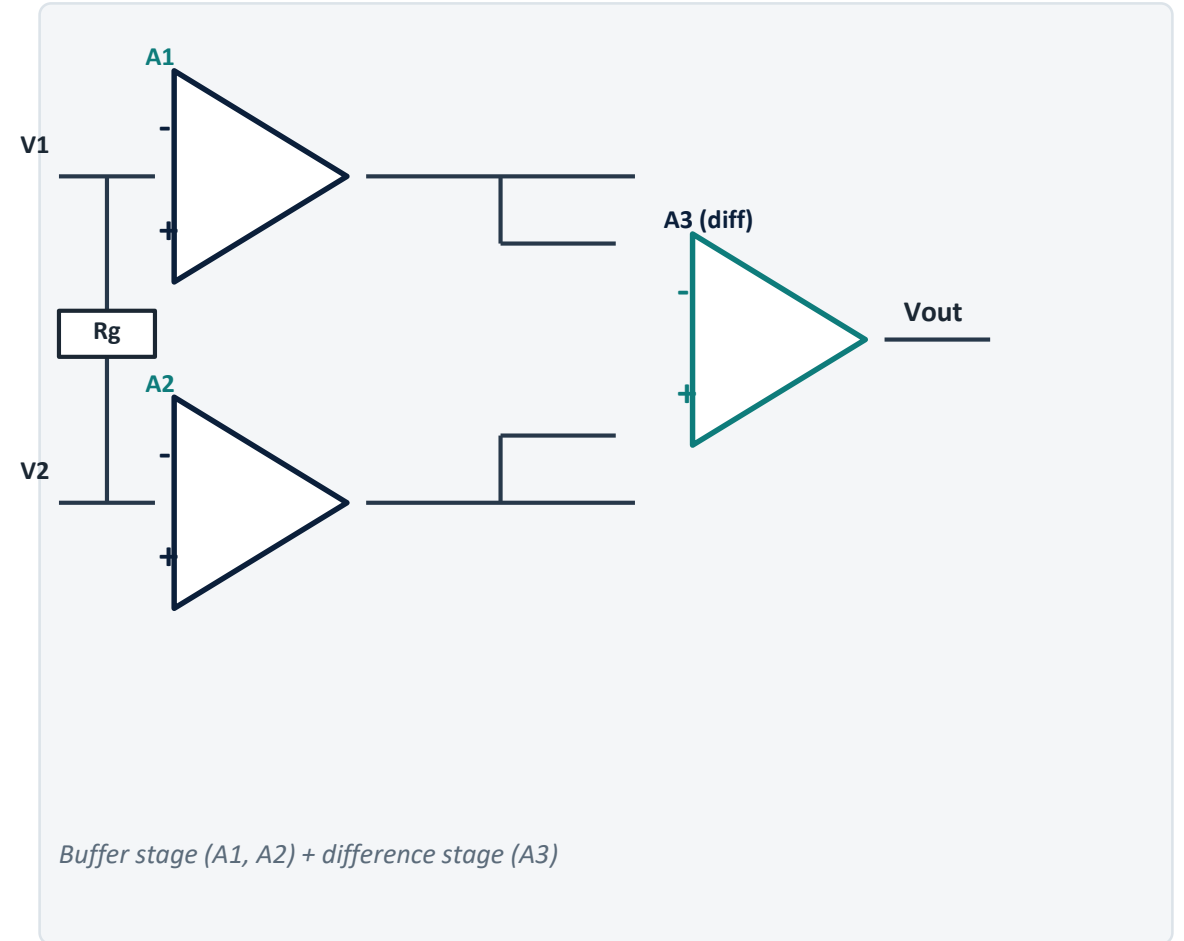


Instrumentation Amplifier

- Three-op-amp design: two buffer/gain stages (A1, A2) followed by a difference amplifier stage (A3)
- Input buffers give very high input impedance and allow gain adjustment using a single resistor (R_g) without affecting CMRR
- Excellent CMRR, high input impedance, precise and stable gain — ideal for amplifying weak signals from sensors, bridges, and transducers (e.g. ECG, strain gauges)
- Overall gain is set by a single external resistor R_g , keeping the resistor network symmetric

GAIN (STAGE 1+2)

$$A_v = (1 + 2R/R_g) \times (R_4/R_3)$$

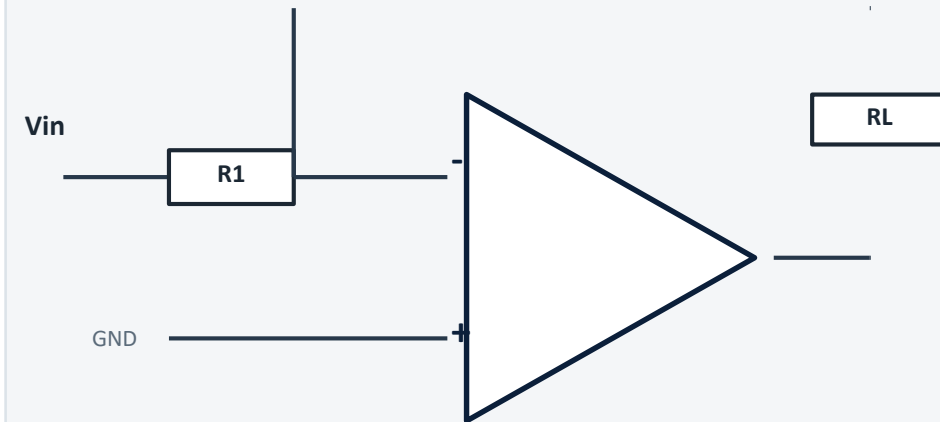


Voltage-to-Current Converter

- Converts an input voltage into an output current that is independent of the load resistance (as long as the op-amp doesn't saturate)
- Floating-load configuration: load is placed in the feedback path; current through it is set entirely by V_{in} and $R1$
- Grounded-load (Howland) configuration: uses matched resistor pairs so the load can be referenced to ground
- Applications: driving LEDs/actuators, calibrating meters, biasing transducers with precise current

LOAD CURRENT (Floating Load)

$$I_L = V_{in}/R1 \quad (\text{independent of } R_L)$$



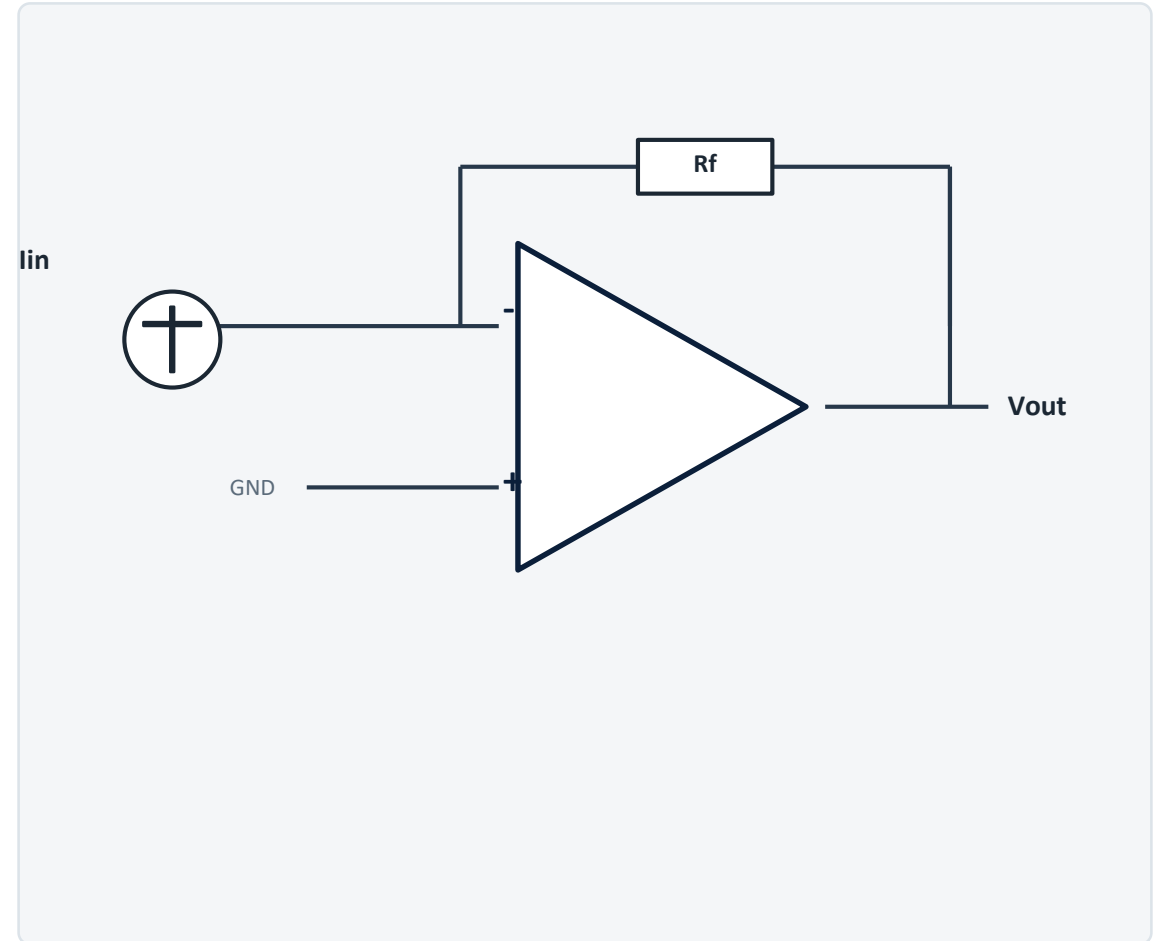
Load R_L sits in the feedback loop — current, not voltage, is controlled

Current-to-Voltage Converter

- Converts an input current (from a photodiode, current-output DAC, or transducer) into a proportional output voltage
- Input current flows directly into the virtual-ground summing junction, so it develops no voltage across the source — ideal for low-impedance current sources
- Feedback resistor R_f sets the conversion (transresistance) gain
- Widely used at the front end of photodiode receivers, current-output sensors and DACs

TRANSRESISTANCE GAIN

$$V_{out} = -I_{in} \times R_f$$

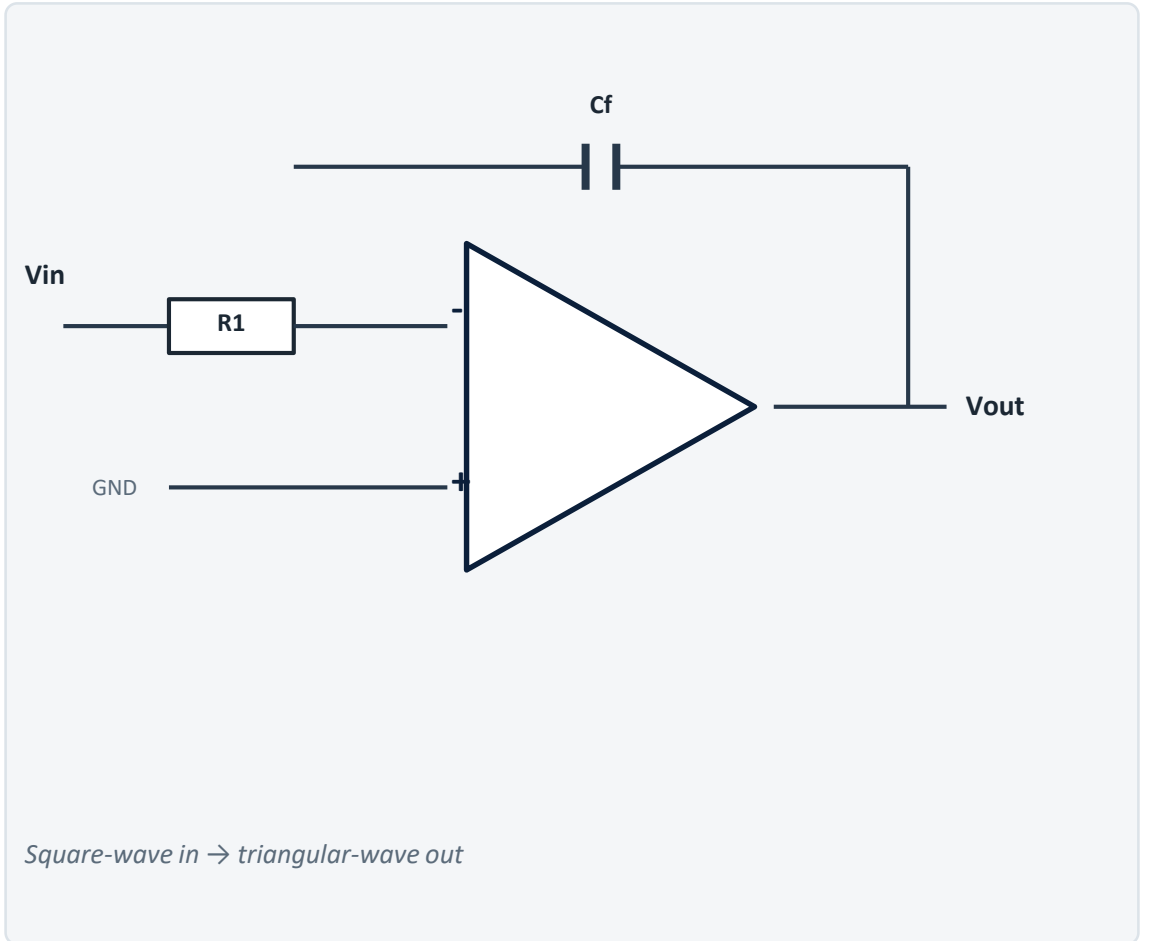


The Integrator

- Feedback element R_f is replaced with a capacitor C_f
- Output is proportional to the negative integral (running area) of the input signal over time
- A square-wave input produces a triangular-wave output
- Practical circuits add a large resistor across C_f to limit DC gain and prevent output drift/saturation

OUTPUT

$$V_{out}(t) = -(1/R1 \cdot C_f) \int V_{in} dt$$

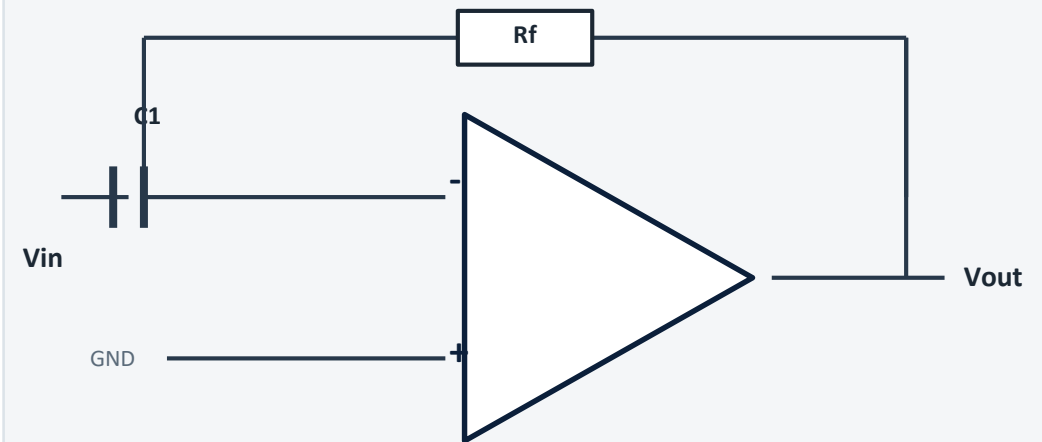


The Differentiator

- Input resistor R1 is replaced with a capacitor C1; Rf remains the feedback resistor
- Output is proportional to the negative rate of change (derivative) of the input signal
- A triangular-wave input produces a square-wave output
- Highly sensitive to high-frequency noise — a small series resistor is often added ahead of C1 to stabilize the circuit

OUTPUT

$$V_{out}(t) = -R_f \cdot C_1 \cdot d(V_{in})/dt$$



Triangular-wave in → square-wave out

02

SECTION TWO

Active Filters & Oscillators

First-Order Low-Pass & High-Pass Filters

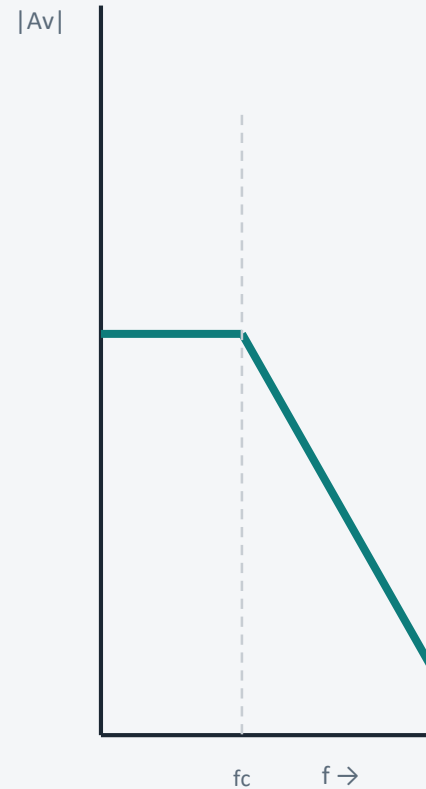
- Built from a single RC network ahead of a non-inverting amplifier stage
- Low-Pass: RC network to ground at the (+) input passes low frequencies, attenuates high frequencies above f_c at -20 dB/decade
- High-Pass: capacitor and resistor swapped — passes high frequencies, attenuates below f_c at -20 dB/decade
- The op-amp stage provides gain/buffering; the passive RC network sets the cutoff

CUTOFF FREQUENCY

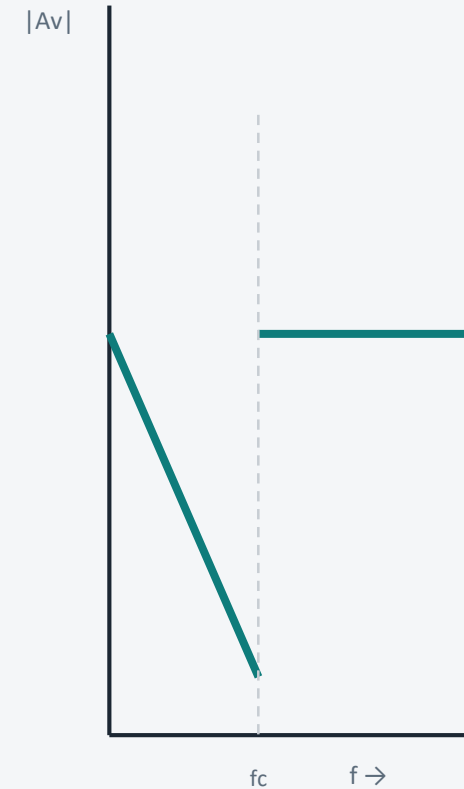
$$f_c = 1 / (2\pi RC)$$

$$\text{Passband gain } A_v = 1 + R_f/R_1$$

LOW-PASS



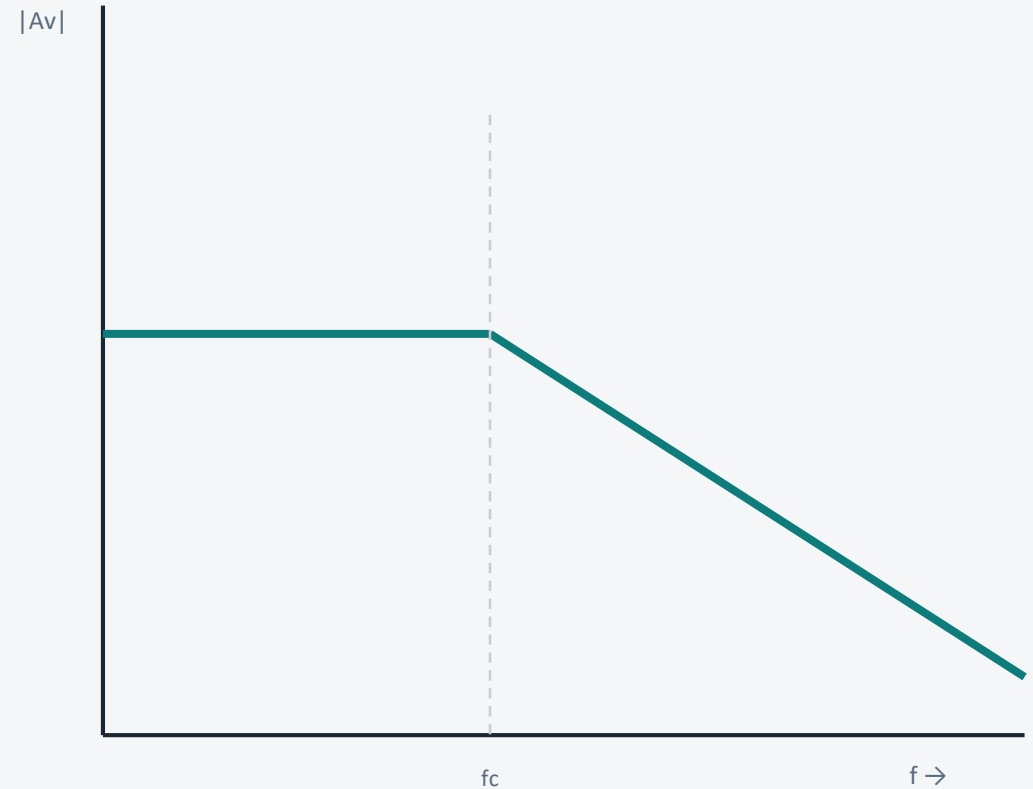
HIGH-PASS



Second-Order Low-Pass & High-Pass Filters

- Uses two RC stages (typically Sallen-Key topology) instead of one, giving a steeper roll-off of -40 dB/decade
- Provides a sharper transition between passband and stopband than a first-order filter
- Filter response shape (Butterworth, Chebyshev, Bessel) is set by the ratio of the resistors/capacitors and the resulting damping factor
- Second-order low-pass: both RC sections placed in the forward path with feedback taken after the second stage
- Second-order high-pass: same topology with resistors and capacitors interchanged
- Higher-order filters (4th, 6th...) are built by cascading second-order sections

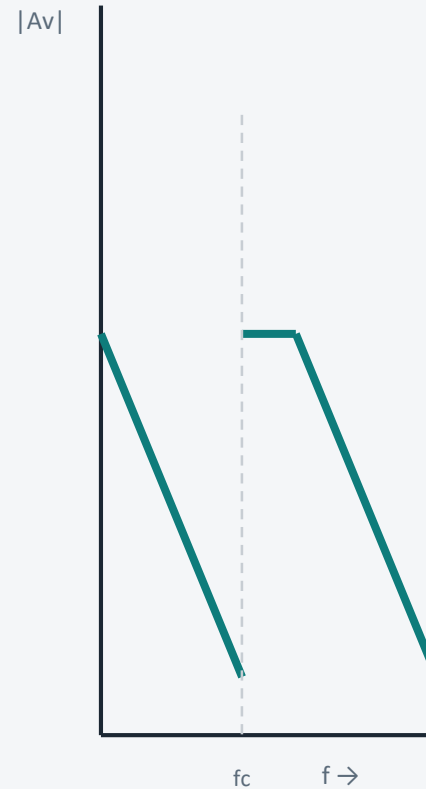
STEEPER ROLL-OFF (-40 dB/DEC)



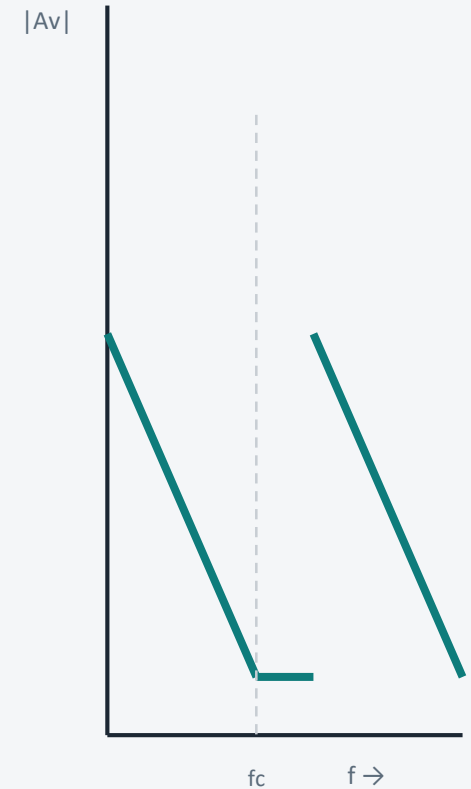
Band-Pass & Band-Reject Filters

- Band-Pass Filter (BPF): cascade of a high-pass stage (sets lower cutoff f_L) and a low-pass stage (sets upper cutoff f_H) — passes only the band between f_L and f_H
- Wide-band BPF: $f_H \gg f_L$, built from separate 1st-order stages
- Narrow-band BPF: uses a multiple-feedback topology for a sharp peak at a single centre frequency f_0 with high Q
- Band-Reject / Notch Filter (BRF): combines a low-pass and high-pass output (parallel paths) summed together to reject a narrow band while passing frequencies on either side
- Widely used to eliminate a specific interference frequency (e.g. 50/60 Hz mains hum)

BAND-PASS



BAND-REJECT (NOTCH)



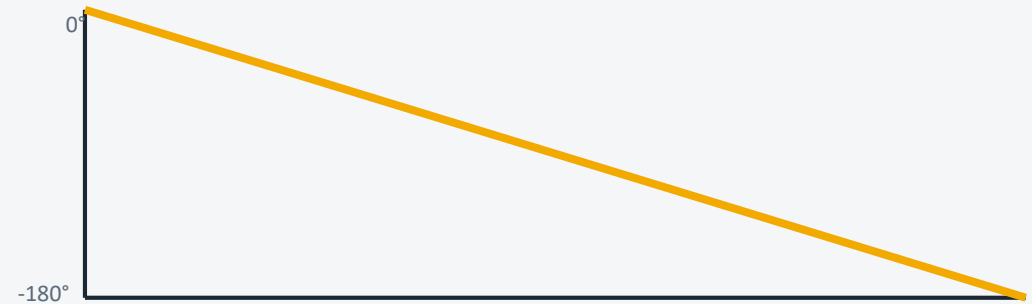
All-Pass Filter

- Passes all frequencies with constant (unity) gain — magnitude response is flat across the spectrum
- Purpose is not to shape amplitude but to shift phase by a controlled, frequency-dependent amount
- Built using an op-amp with equal resistors R in the gain-setting network and an RC network at the input that produces the phase shift
- Phase shift ranges from 0° to -180° (lag configuration) or 0° to $+180^\circ$ (lead configuration) as frequency increases
- Used for phase correction/equalization and to build phase-shift oscillators

FLAT MAGNITUDE RESPONSE

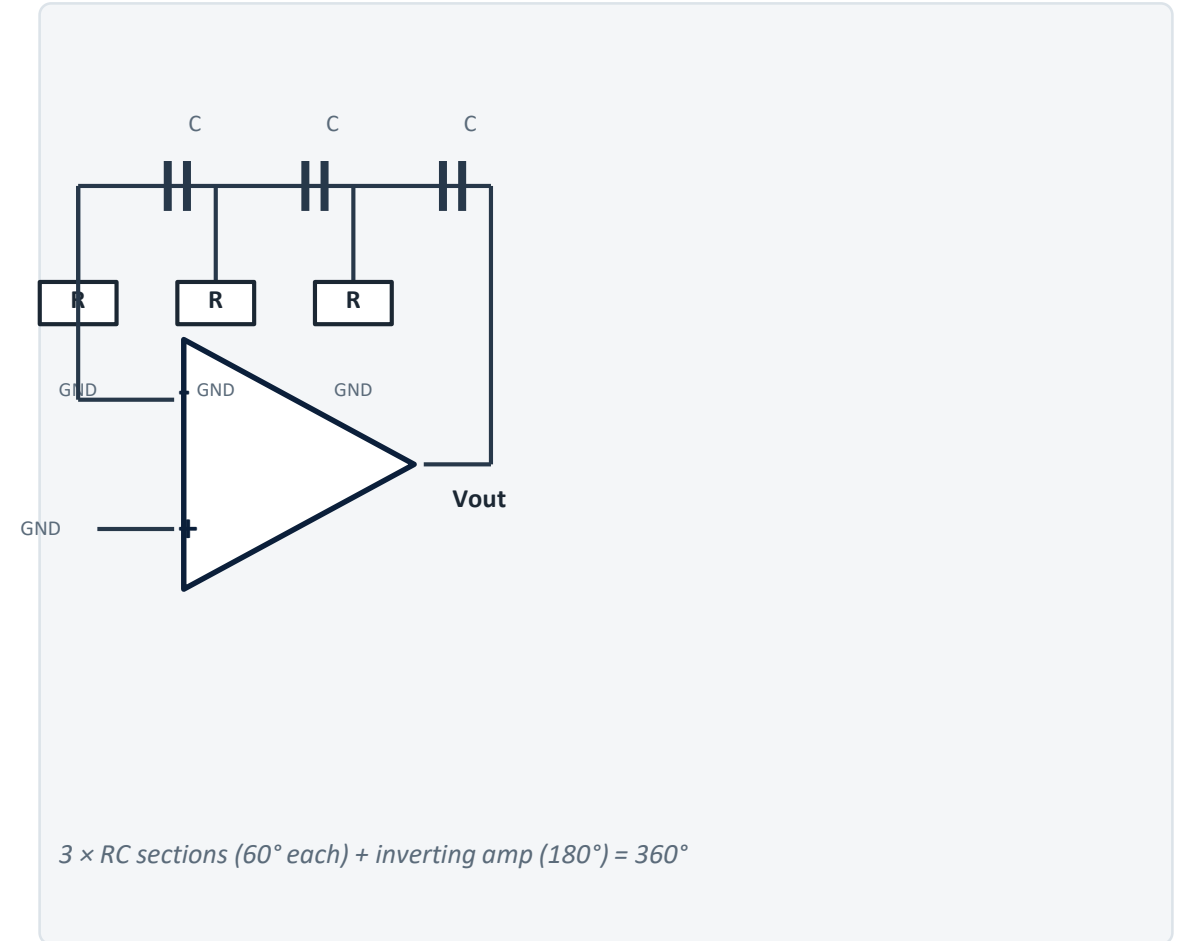


PHASE RESPONSE



RC Phase-Shift Oscillator

- Three cascaded RC sections in the feedback path, each contributing 60° of phase shift at the oscillation frequency — together giving the required 180°
- The inverting op-amp stage adds the remaining 180°, satisfying the Barkhausen criterion: total phase shift = 360°
- Loop gain must equal 1 at f_o — requires the amplifier gain to be at least 29 to overcome the RC network's attenuation
- Produces a reasonably good sine wave; simple but has moderate frequency stability



OSCILLATION FREQUENCY

$$f_o = 1 / (2\pi RC\sqrt{6})$$

$$\text{Minimum gain } |A_v| = 29$$

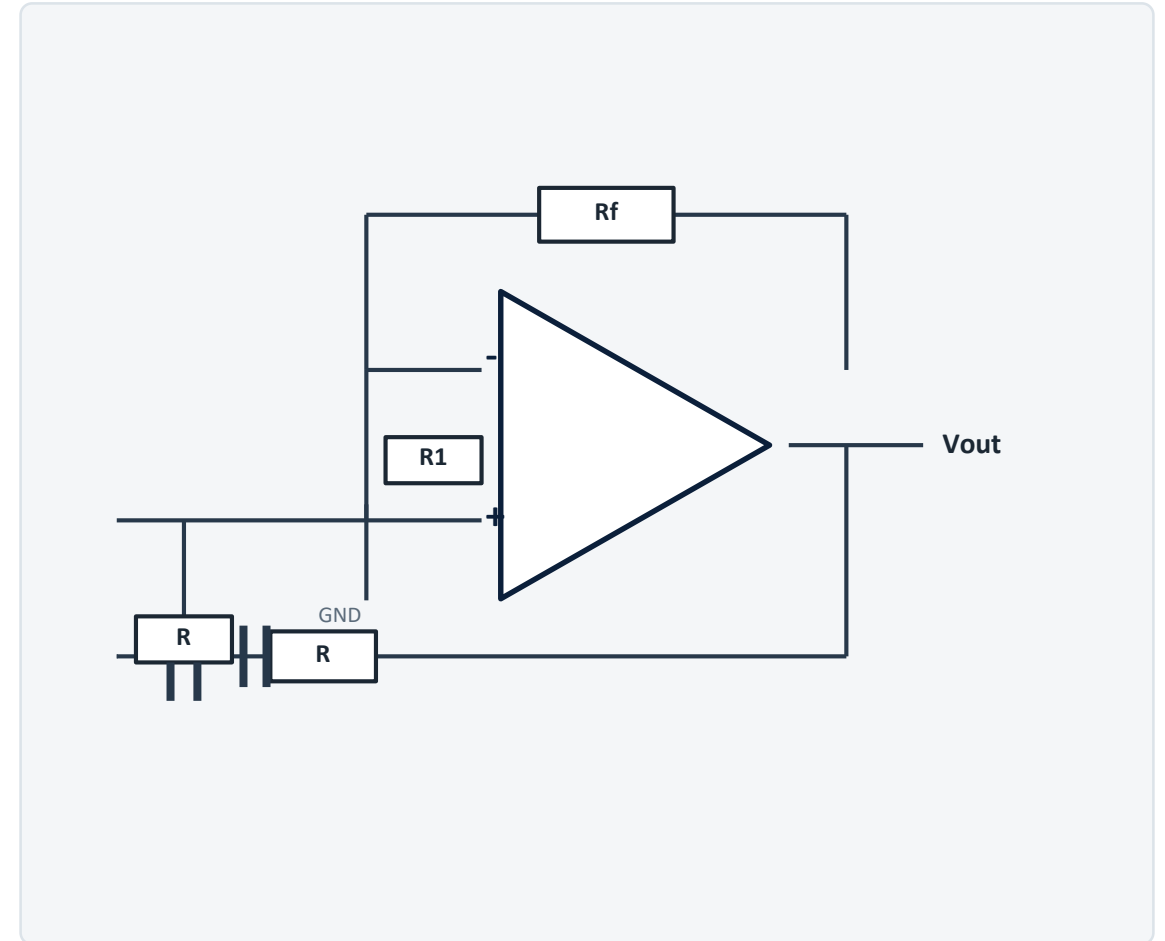
Wien Bridge Oscillator

- Uses a lead-lag RC network (series RC + parallel RC) in the positive feedback path, which gives zero phase shift and maximum attenuation of 1/3 at the oscillation frequency
- Op-amp is configured as a non-inverting amplifier with gain of exactly 3 to overcome the network's attenuation
- Amplitude stabilization is typically achieved with a small lamp or diode-limiter in the gain-setting feedback resistor
- Produces a very pure, low-distortion sine wave — the most common audio-frequency oscillator

OSCILLATION FREQUENCY

$$f_o = 1 / (2\pi RC)$$

$$\text{Required gain } A_v = 3 \quad (R_f = 2R_1)$$

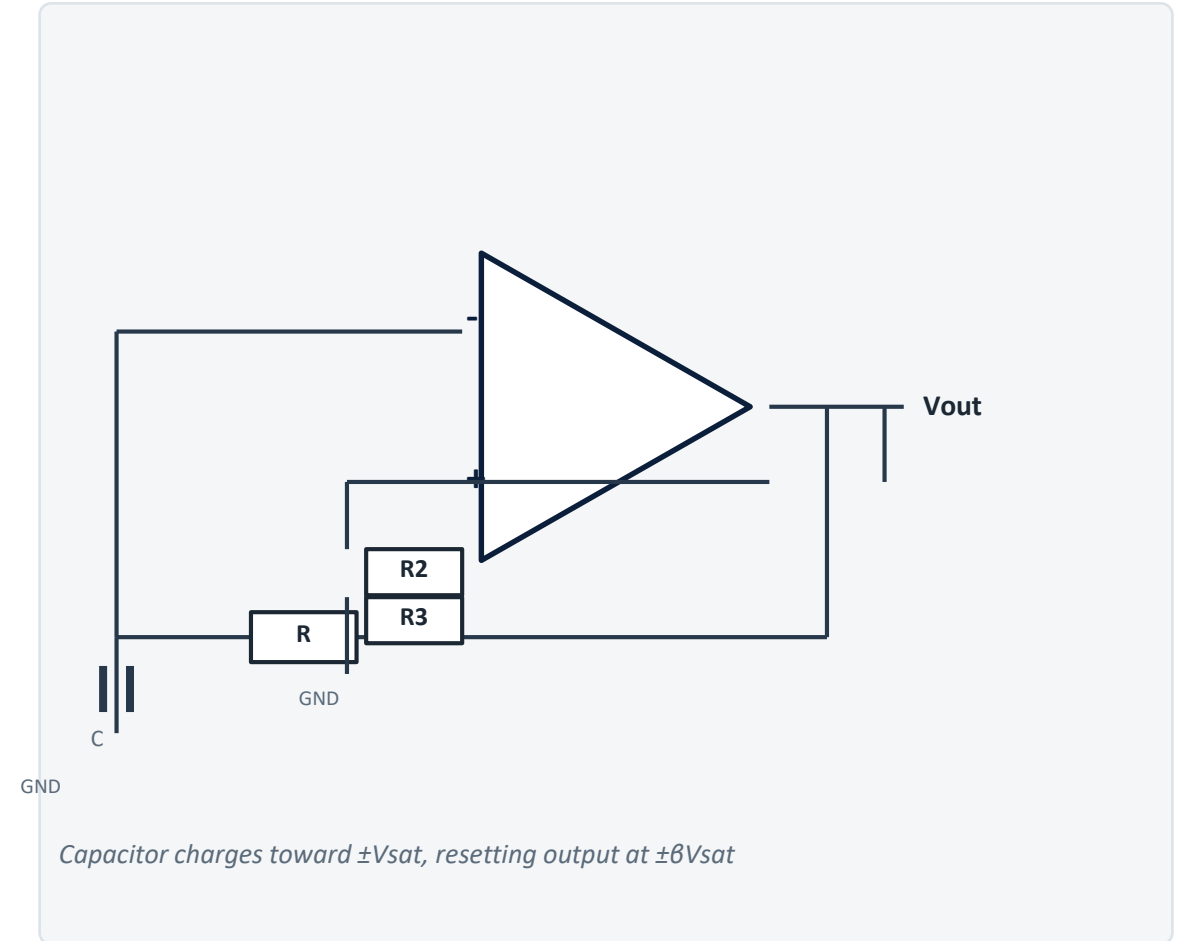


Square Wave Generator (Astable)

- Op-amp used as a free-running comparator with positive feedback through a resistive divider (sets threshold $\pm\beta V_{sat}$)
- A capacitor at the (-) input charges/discharges through resistor R toward the output saturation voltage
- Each time the capacitor voltage crosses the threshold, the output flips between $+V_{sat}$ and $-V_{sat}$, recharging the capacitor in the opposite direction
- Output is a symmetric square wave whose frequency is set by R, C and the feedback ratio β

FREQUENCY ($R_2=R_3$, $\beta=0.5$)

$$f_o = 1 / (2RC \cdot \ln(3))$$



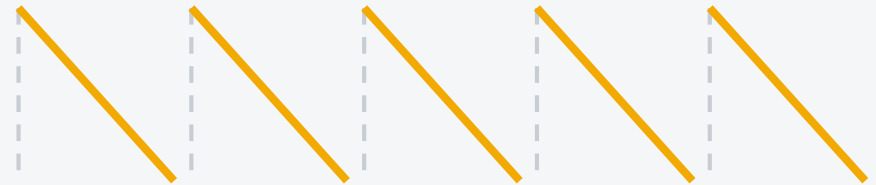
Triangular & Sawtooth Wave Generators

- Triangular generator: cascades a square-wave generator (comparator) with an integrator — the comparator's square output is integrated to form a linear triangular wave
- The integrator's output is fed back to the comparator's threshold input, so the two stages lock together at the same frequency
- Amplitude of the triangle wave is set by the comparator's saturation levels and the RC time constant of the integrator
- Sawtooth generator: modifies the triangular generator by making the charge and discharge times unequal — e.g. using a diode in parallel with the integrator resistor so charging is fast and discharging is slow (or vice-versa)
- This asymmetry produces a ramp that rises (or falls) linearly then resets sharply

TRIANGULAR OUTPUT



SAWTOOTH OUTPUT





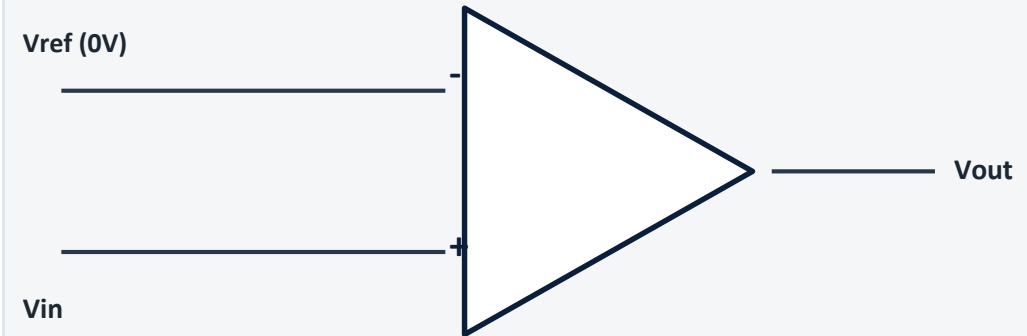
03

SECTION THREE

Comparators & Converters

Basic Comparator & Zero-Crossing Detector

- Comparator: op-amp used open-loop (no feedback) — output swings fully to $+V_{sat}$ or $-V_{sat}$ depending on which input is larger
- Non-inverting comparator: input to (+), reference to (-); output goes high when $V_{in} > V_{ref}$
- Inverting comparator: input to (-), reference to (+); output goes low when $V_{in} > V_{ref}$
- Zero-Crossing Detector: special case with $V_{ref} = 0\text{ V}$ — output switches state precisely every time the input signal crosses zero
- Used to convert a sine/analog waveform into a synchronized square wave, and to measure frequency or trigger timing circuits



No feedback — output slams to $\pm V_{sat}$ based on sign of $(V_{in} - V_{ref})$

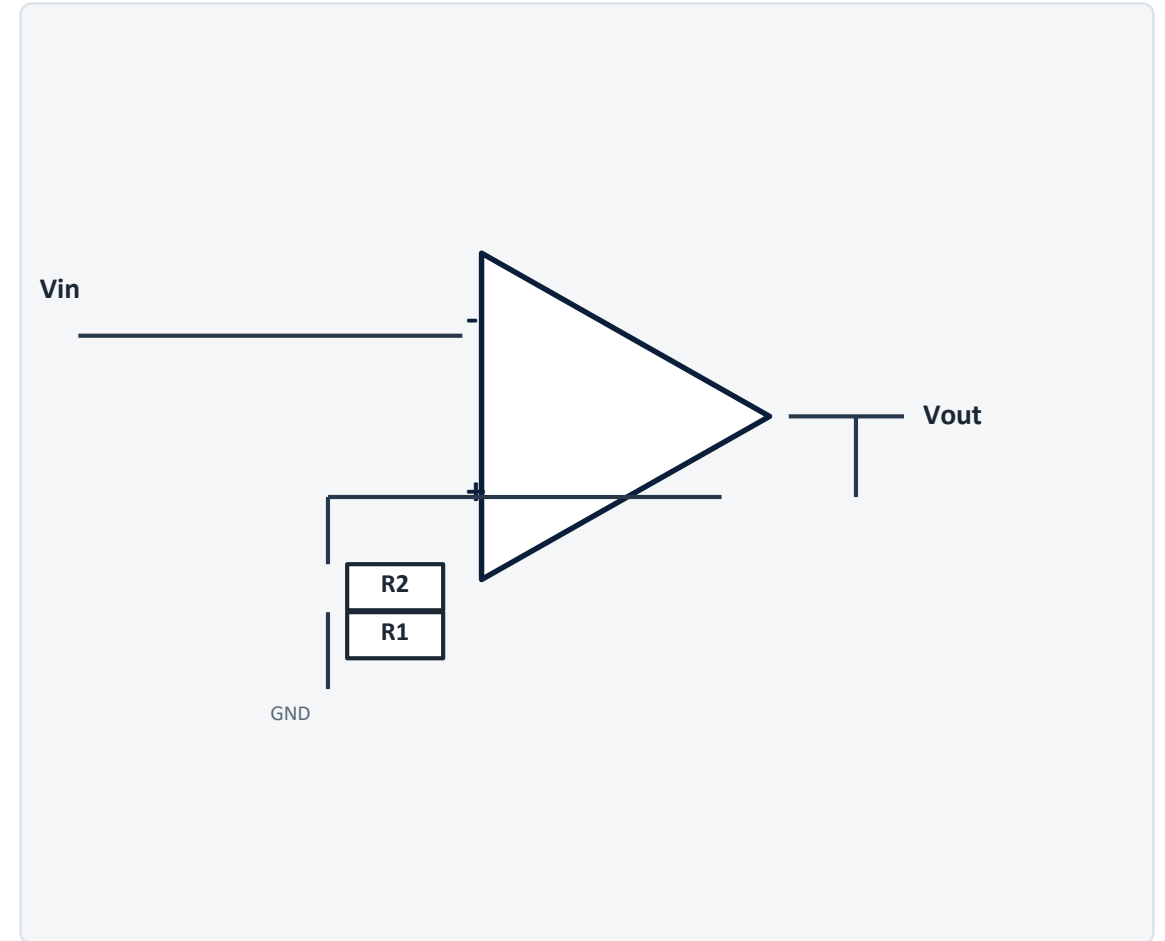
Schmitt Trigger

- A regenerative comparator with positive feedback through a resistive divider (R1, R2) to the (+) input
- Creates two distinct switching thresholds — Upper Threshold Point (UTP) and Lower Threshold Point (LTP) — instead of one
- This hysteresis makes the circuit immune to noise near the switching point, unlike a plain comparator
- Commonly used to convert a slow-changing or noisy analog signal into a clean, jitter-free square wave

THRESHOLDS

$$UTP = +V_{sat} \cdot R1 / (R1 + R2)$$

$$LTP = -V_{sat} \cdot R1 / (R1 + R2)$$



Clippers & Clampers

- Positive Clipper: removes (flattens) the positive portion of the waveform above a set reference level, using a diode in the feedback path that conducts only above that level
- Negative Clipper: same principle, removes the negative portion of the waveform below a reference level
- Clippers are precision circuits — the op-amp's gain compensates for the diode's forward-voltage drop, giving a sharp, accurate clipping level unlike a simple diode clipper
- Clamper: shifts the entire waveform up or down by a DC level without changing its shape, using a diode-capacitor network so the peak (or trough) is clamped to a reference voltage
- Positive clamper shifts the waveform upward (lowest point clamped to V_{ref}); negative clamper shifts it downward

CLIPPED WAVEFORM



CLAMPED WAVEFORM (SHIFTED UP)



Absolute-Value Circuit & Peak Detector

- Absolute-Value Circuit (Precision Rectifier): combines a half-wave precision rectifier stage with a summing/inverting stage so both halves of the input waveform appear as positive output — effectively a full-wave rectifier with no diode voltage-drop error
- Output is always $|V_{in}|$, tracking the input magnitude regardless of its original polarity
- Peak Detector: a diode in the feedback path charges a storage capacitor to the highest voltage the input has reached; the capacitor holds that peak value at the output
- The op-amp's high input impedance and buffering prevent the capacitor from discharging back through the source
- A reset switch (or bleed resistor) across the capacitor is used to clear the stored peak for a new measurement cycle

$|V_{in}|$ OUTPUT

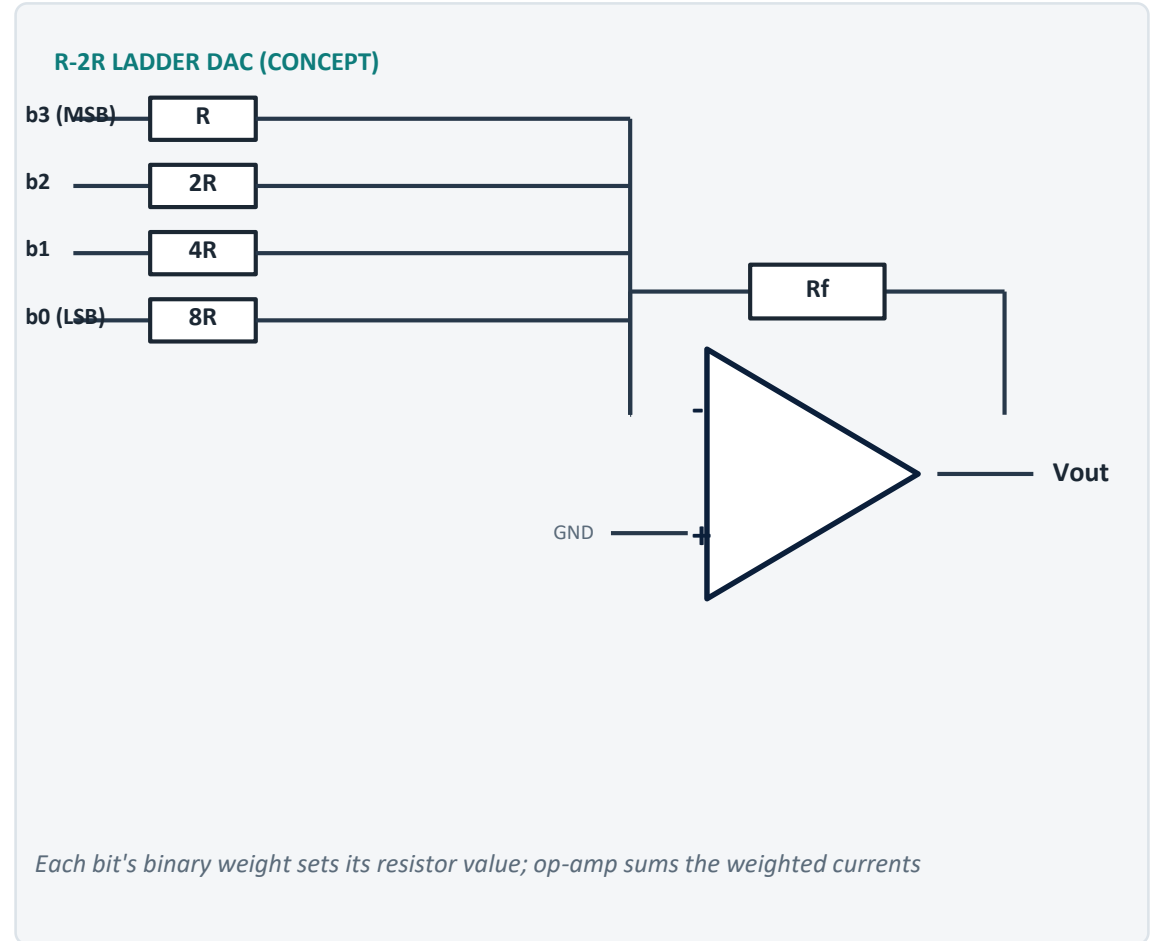


PEAK-HOLD OUTPUT



ADC and DAC using Op-Amp

- DAC (Digital-to-Analog Converter): a binary-weighted resistor network (or R-2R ladder) feeds a summing amplifier — each bit contributes a current proportional to its weight, and the op-amp sums them into an analog output voltage
- R-2R ladder DAC is preferred in practice since it needs only two resistor values, giving better accuracy and easier fabrication than binary-weighted resistors
- ADC (Analog-to-Digital Converter): op-amp comparators form the core decision elements
- Flash ADC: a bank of comparators (one per quantization level) compares V_{in} against reference voltages simultaneously — fastest type, but needs many comparators
- Successive-Approximation ADC: uses one comparator with a DAC in the feedback loop, testing one bit at a time from MSB to LSB — slower but far fewer components



SUMMARY

Op-Amps: One Building Block, Countless Applications

Linear Circuits

- Amplifiers & followers
- Summing / scaling / averaging
- Differential & instrumentation
- V-I, I-V, integrator, differentiator

Filters & Oscillators

- 1st & 2nd order LP/HP filters
- Band-pass, band-reject, all-pass
- Phase-shift & Wien bridge
- Square, triangular, sawtooth gen.

Comparators & Converters

- Comparator & zero-crossing
- Schmitt trigger (hysteresis)
- Clippers, clampers, abs-value, peak
- ADC & DAC using op-amps